MIL-R-14864(MU)
AMENDMENT 2
30 May 1972
SUPERSEDING
AMENDMENT 1
24 March 1972

MILITARY SPECIFICATION

RANGE FINDER, LASER AN/VVS-1

This amendment forms a part of Military Specification MIL-R-14864(MU), dated 2 March 1970.

* Page 3: Add paragraph 3.1.1 as follows:

"3.1.1 Interchangeability.- The range finder shall meet all of the following requirements of this specification without need for selective assembly of the units and items listed in a through i of 3.1. They shall be interchangeable such that replacement of any single item or random combination of items of the system permits the system to operate and perform to the requirements of this specification. No incompatibility shall exist between the item replacement and the mechanical interface/fits or electrical functions of the range finder."

Page 3 - Table I, Item 1.12: Delete "240" and substitute "1800" and for Item 1.13: Delete "620" and substitute "2700".

Page 8: Renumber the following paragraphs:

FROM	TO
" 3.3.2.1	3.3.2.1.1
3.3.2.2	3.3.2.2.1
3.3.2.3	3.3.2.2.2 "

Page 8: Add the following new paragraphs:

* "3.3.2.1 Non-operating conditions.- The laser range finder system shall sustain without damage or deterioration of performance the following environmental exposure when in a non-operating (de-energized) condition.

* 3.3.2.1.1 Storage temperature.- The range finder shall meet the requirements of 3.3.1.3 (except subparagraph (c)(4)), 3.3.1.6.1 (max range only) and 3.3.1.11 at ambient temperature (60°F to 90°F) after exposure and thermal stabilization at -80°F and +160°F.

* 3.3.2.1.2 Humidity.- The laser range finder system shall meet the requirements of MIL-STD-810, Method 507, Procedure I after which, the range finder system shall meet the requirements of 3.3.1.3 (except subparagraph (c) (4)), 3.3.1.6.1 (max. range only) and shall accomplish a minimum of 200 rangings in accordance with the reliability demonstration test, para. 4.6.2.7.

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* 3.3.2.1.3 Salt fog.- The laser range finder system shall meet the requirements of MIL-STD-810, Method 509 after which, the range finder system shall meet the requirements of 3.3.1.3 (except subparagraph (c)(4)), 3.3.1.6.1 (max. range only) and shall accomplish a minimum of 200 rangings in accordance with the reliability demonstration test, para. 4.6.2.7.

* 3.3.2.1.4 Dust.- The laser range finder system shall meet the requirements of MIL-STD-810, Method 510 after which, the range finder system shall meet the requirements of 3.3.1.3 (except subparagraph (c)(4)) 3.3.1.6.1 (max range only) and shall accomplish a minimum of 200 rangings in accordance with the reliability demonstration test, para. 4.6.2.7.

* 3.3.2.1.5 Transportability (shock).- The laser range finder system shall meet the requirements of MIL-STD-810, Method 516, Procedure II after which, the range finder system shall meet the requirements of 3.3.1.3 (except subparagraph (c)(4)), 3.3.1.6.1 (max range only) and shall accomplish a minimum of 200 rangings in accordance with the reliability demonstration test, para. 4.6.2.7.

* 3.3.2.1.6 Fungus.- The laser range finder system shall meet the requirements of MIL-STD-810, Method 508 after which the range finder system shall meet the requirements of 3.3.1.3 (except subparagraph (c)(4)), 3.3.1.6.1 (max. range only) and shall accomplish a minimum of 200 rangings in accordance with the reliability demonstration test, para. 4.6.2.7.
"NOTE: The laser range finder system undergoing testing for fungus does not have to be subjected for test to meet other environmental requirements."

* 3.3.2.2 Operating conditions.- The laser range finder system shall operate without damage or deterioration of performance during the following environmentla exposure.

NOTE: Paragraphs 3.3.2.2.1 and 3.3.2.2.2 (operating temperature and vibration) were originally numbered 3.3.2.2 and 3.3.2.3 respectively.

* 3.3.4 Reliability: The range finder shall have a mean ranging between failure (MRBF) of 5000 rangings."

Page 15 and 16, Table V - In the "Range (NIXIES)" column, delete tol. Value (± 5 meters)" and substitute "(±10 meters)."

* Page 9, paragraph 4.2 at end of 2nd sentence, add: "plus 3 each of all items covered by SQAP."

* Page 10, paragraph 4.3.2.3, lines 4 and 7: Add Table "IV-B".

* Page 11, Table IV:

Characteristic 304: Change "3.3.2.2" to read "3.3.2.2.1"
Characteristic 305: Change "3.3.2.3" to read "3.3.2.2.2"
Add Characteristic "306" as follows:
"306. Interchangeability 3.3.1 4.6.2.3"

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* Page 11: Add Tables IV-A and IV-B as follows:

TABLE IV-A

	CONTROL TEST	REQUIREMENT	TEST PROCEDURE
307.	Salt Fog	3.3.2.1.3	4.6.2.4
308.	Dust	3.3.2.1.4	4.6.2.5
309.	Fungus	3.3.2.1.6	4.6.2.6
310.	Reliability	3.3.4	4.6.2.7

TABLE IV-B

	CONTROL TEST	REQUIREMENT	TEST PROCEDURE
311.	Humidity	3.3.2.1.2	4.6.2.8
312.	Transportability	3.3.2.1.5	4.6.2.9

* Page 12, paragraph"4.4.3": Renumber to "4.4.5"

* Page 12: Add paragraphs 4.4.3 and 4.4.4 as follows:

"4.4.3 First Article Testing.- One range finder of the first article sample shall be subjected to all of the tests in Tables II, III, Interchangeability Test of Table IV, and fungus and reliability tests of Table IV-A; Two range finders shall be subjected to all of the tests of Tables II, III, IV, IV-A and IV-B, except fungus test of Table IV-A, to determine compliance with 4.2. All examinations and tests shall be witnessed by the Government Quality Assurance Representative. When manufacturing processes, materials, or inspection techniques previously applied are subject to change, or an adverse change occurs in performance capability or reliability of the range finder, this shall be cause to cease those operations contributing to the change or failure until corrections have been made by the contractor and approved by the Government.

4.4.4 Special environment.- One range finder shall be randomly selected by a Government representative at 25%, 50% and 75% completion of the contract production and shall meet the requirements and tests of Table IV-B.

* Page 20, add paragraphs 4.6.2.3 to 4.6.2.9 as follows:

* "4.6.2.3 Interchangeability.- The Government Quality Assurance Representative shall select units and items listed in 3.1 for final assembly of 3 range finders by the contractor to determine compliance with 3.3.1 and 3.3.3. Selection shall be as follows:

a. Units and items required to assemble the first range finder shall be randomly selected from a quantity of units and items sufficient to assemble 3 range finders.

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b. Units and items required to assembly the second range finder shall be randomly selected from remaining units and items.

c. The third range finder shall be assembled from remaining units and items.

NOTE: Subsequent to final assembly of each range finder, adjustment of PFN voltage, permitted if necessary, will be permitted to obtain the specified output.

4.6.2.4 Salt fog.- Perform this test utilizing the apparatus (chamber atomizers, air supply, and salt solution) specified in Method 509 of MIL-STD-810, in accordance with Procedure I therein, except the range finder shall not be operated during test. The range finder shall meet the requirements of 3.3.2.1.3 upon completion of test.

* 4.6.2.5 Dust.- Perform this test utilizing the apparatus specified in Method 510 of MIL-STD-810, in accordance with Procedure I therein. The range finder shall not be operated during performance of tests (Steps 1, 2 or 3). Step 3 may be performed after reaching stabilization in Step 2. Window of Receiver-Transmitter shall be thoroughly cleaned prior to Lasing Operations. The range finder shall meet the requirements of 3.3.2.1.4 upon completion of test.

* 4.6.2.6 Fungus.- Perform this test utilizing the apparatus specified in Method 508 of MIL-STD-810, in accordance with Procedure I therein, with the unit non-operating during test. Range finder shall meet the requirements of 3.3.2.1.6 upon completion of test.

* 4.6.2.7 Reliability.- A reliability demonstration test shall be conducted as a part of First Article Testing to assure the requirement of para. (3.3.4) at a 90% confidence level. A minimum of 200 rangings on each sample range finder shall be accumulated after/during exposure to the non-operating/operating environments paragraph 3.3.2. Additional rangings are required at 75 ± 15°F in order to accumulate a minimum of 3000 rangings on each sample range finder. The sample range finders shall be tested until the total rangings (11,500 minimum) satisfies the following relationship or until 6 failures have occurred or 47,000 rangings (maximum) have been accomplished:

$$5000 \leq \frac{2T}{\chi^2_{.90,\ 2\ (r+1)}}$$

Where: T = Total accumulated rangings of sample range finders

$\chi^2_{.90\ \ 2\ (r+1)}$ = Chi Square Lower 90% Confidence Level with $2(r+1)$ degrees of freedom

r = Number of mission failures

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* 4.6.2.7.1 Data collected will consist of all equipment malfunctions, number of rangings completed when malfunction occurred, the nature and cause of the malfunction time required to correct the malfunction, and the required corrective action.

* 4.6.2.7.2 Failure definitions.- Any malfunction which renders the range-finder inoperative or operative outside the limits of this specification for the determination of chargeable reliability failures, the following clarification shall apply:

a. Upon failure analysis (to determine the cause of any malfunction) it is determined that an "assignable cause" (related to: Manufacturing process, procedures basic design weakness, etc.) was the factor which caused the occurance of the malfunction, the malfunction will not be chargeable as a reliability failure. However, the correction to the manufacturing process, or procedure (et al) shall be accomplished and the testing resumed at the environment at which the malfunction occurred.

b. Upon failure analysis no assignable cause is evident and the malfunction is considered random in nature, the malfunction shall be a "chargeable" reliability failure and as such must be considered in the above demonstration equation (4.6.2.7).

4.6.2.8 Humidity.- Perform this test utilizing apparatus specified in Method 507 of MIL-STD-810, in accordance with Procedure I. Reverse steps 7 and 8. Operational tests are to be initiated within one hour after removal of range finder from chamber. Range finder shall meet the requirements of 3.3.2.1.2 (Step 8) except reliability demonstration test (200 rangings) is not required after first article approval.

* 4.6.2.9 Transportability (Shock).- Perform this test in accordance with Method 516 of MIL-STD-810, Procedure II. Range finder shall meet the requirements of 3.3.2.1.5 upon completion of test except reliability demonstration test (200 rangings) is not required after first article approval.

* The margin of this specification is marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue."

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